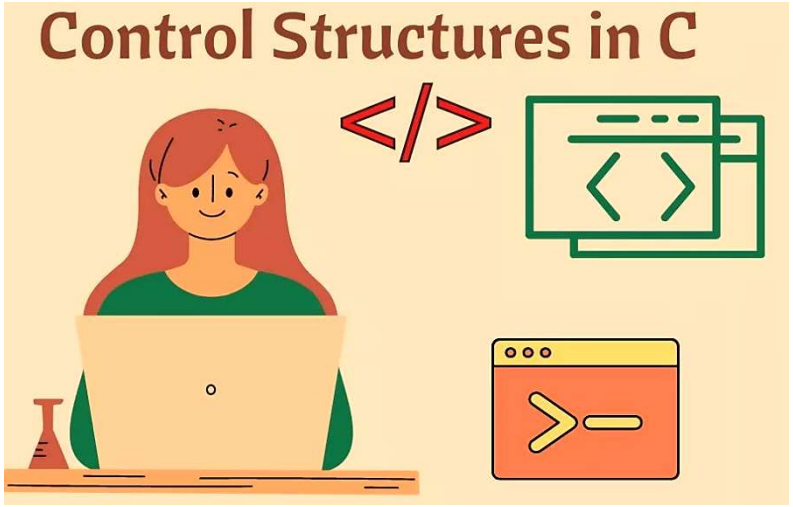
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Structured Programming Concepts

(26CSE0105)



Nested Loops & Pattern Programming in C

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Topics to be covered

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- ***Nested Loops:***
  - ***For loop***
  - ***While loop***
- ***Pattern Programming:***
  - ***Break***
  - ***Continue***
  - ***goto***

Nested Loop Statements

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- *A **nested loop** is a loop **inside another loop**.*
- ***Outer loop:** **Controls the main repetitions**.*
- ***Inner loop:** **Executes completely for each iteration of the outer loop**.*

(If the outer loop runs M times and inner loop runs N times, the inner loop's body will execute a total of M × N times.)

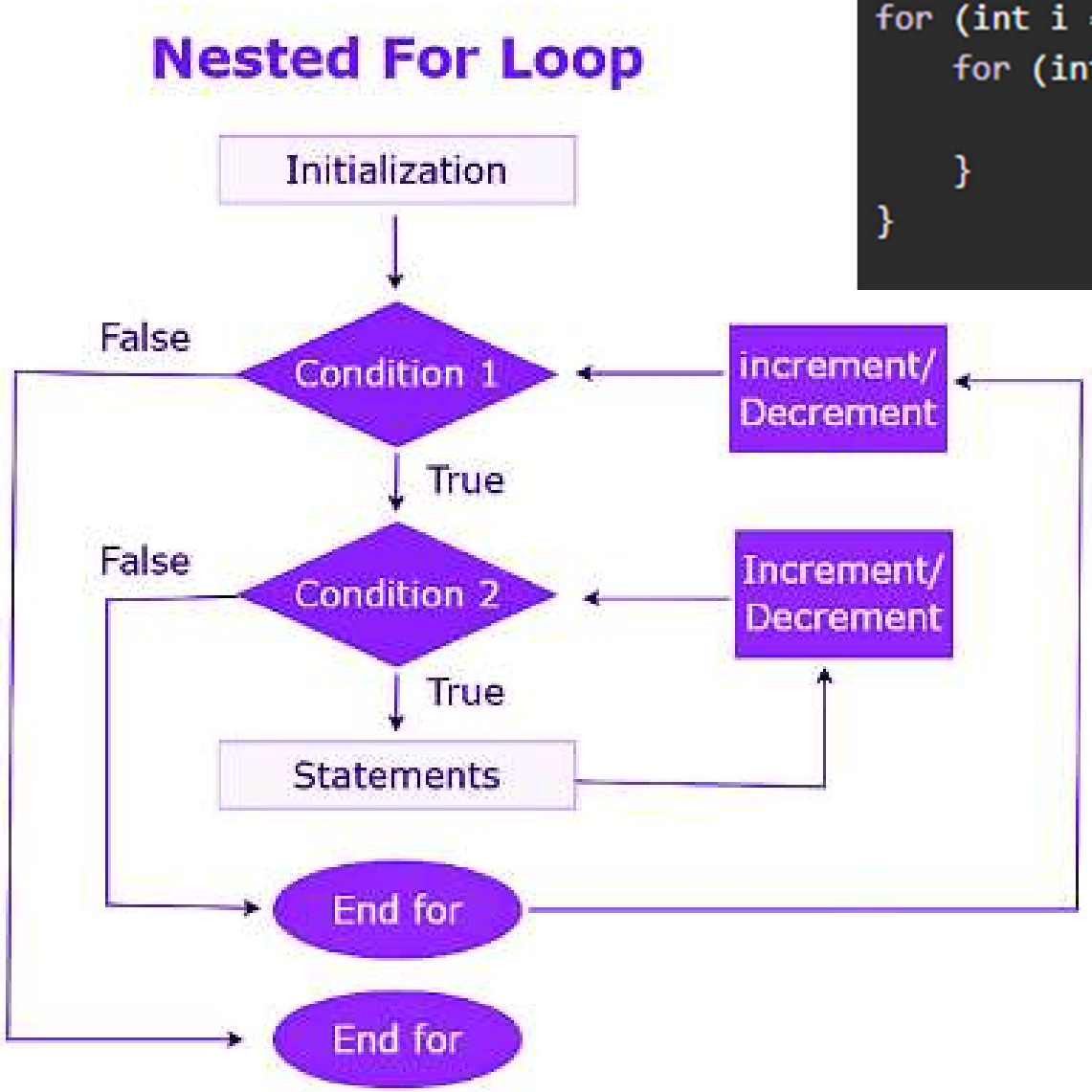
- ***The three main types of nested loops are:***
  - ***Nested For Loop***
  - ***Nested While Loop***
  - ***Nested Do-While Loop***

Nested Loop (for)

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```
for (int i = 0; i < outer_limit; i++) {
    for (int j = 0; j < inner_limit; j++) {

    }
}
```



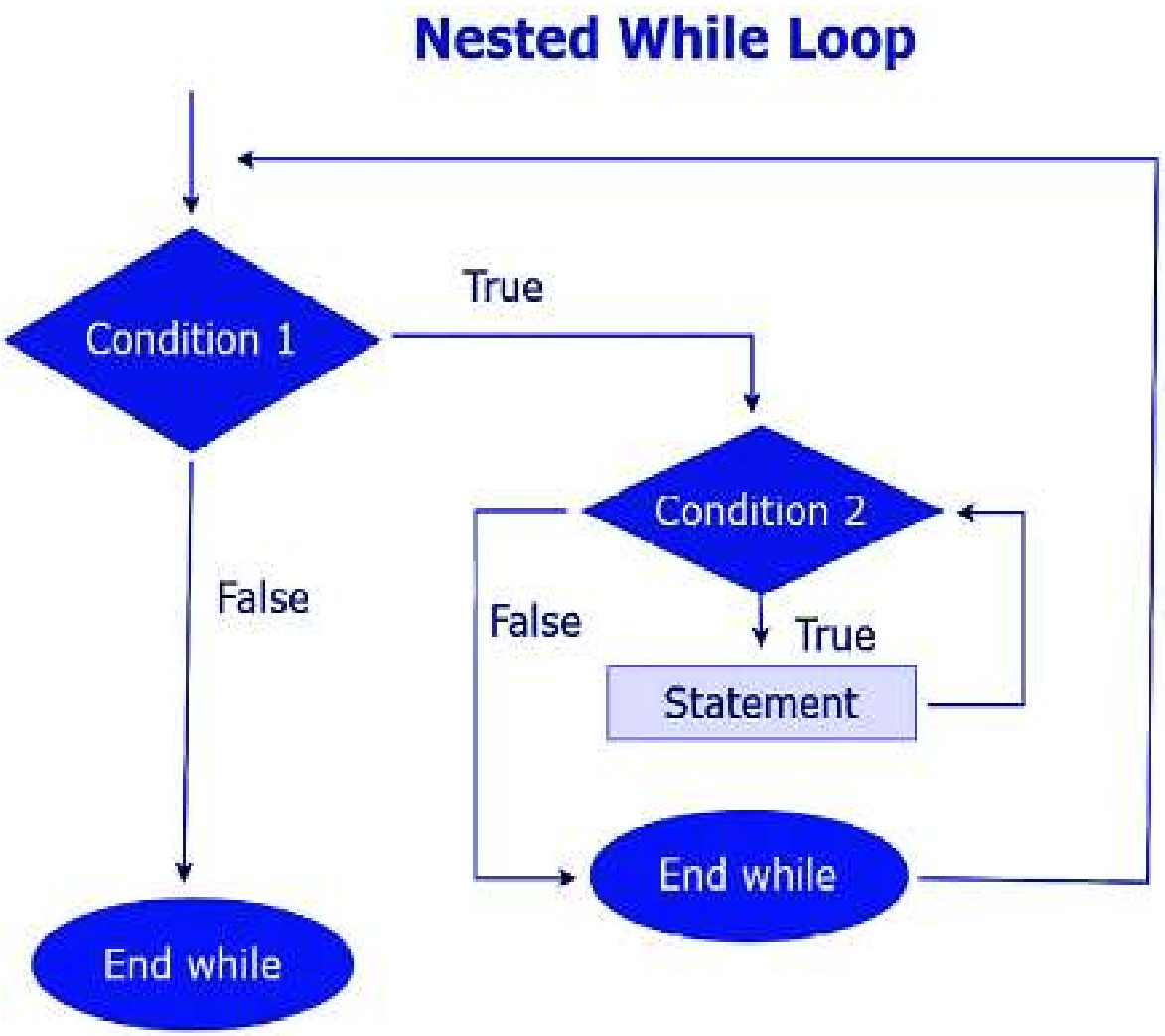
Nested Loop (while)

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```
while (condition) {
    while (condition) {

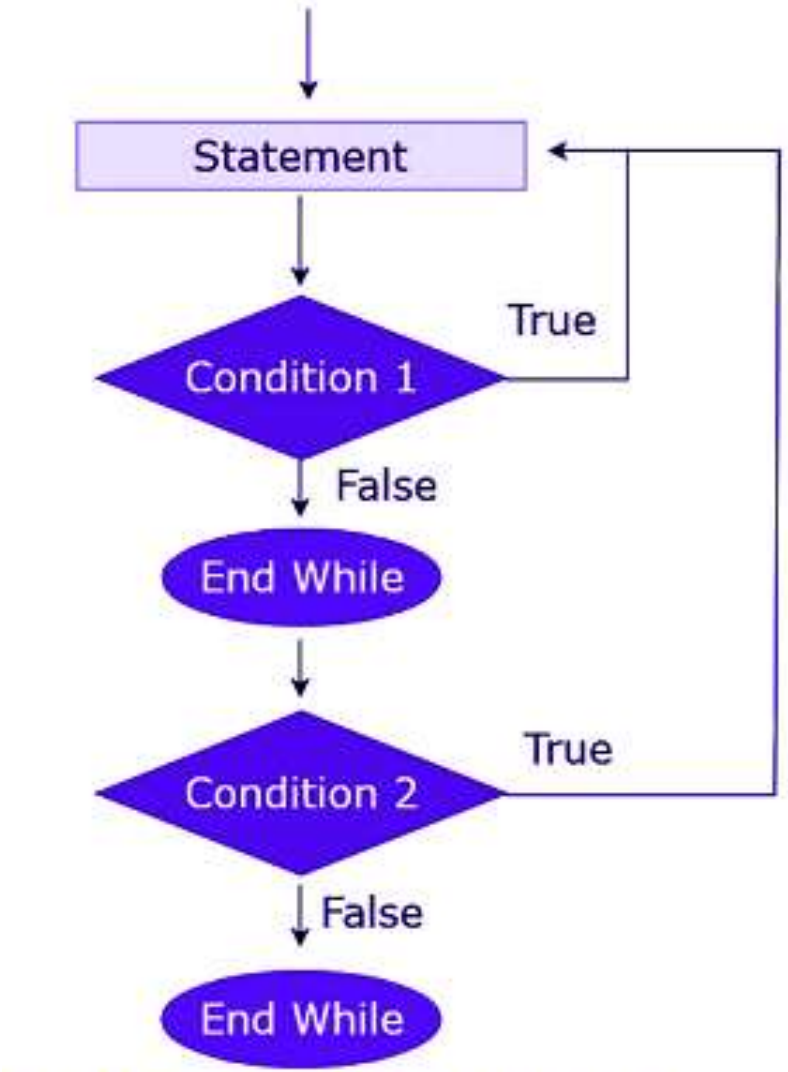
    }

}
```



Nested Loop (do while)

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Nested Do-While Loop

```
do {
    // Outer loop statements

    do {
        // Inner loop statements
    } while (inner_condition);


} while (outer_condition);
```

Nested for loop (example)

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```
#include <stdio.h>
int main()
{
   int rows, cols;
   printf("Enter the number of rows and columns: ");
   scanf("%d%d", &rows, &cols);
   printf("\nMultiplication Table:\n");
   for (int i = 1; i <= rows; i++)
   {
      for (int j = 1; j <= cols; j++)
      {
         printf("%4d", i * j);
      }
      printf("\n");
   }
   return 0;
}
```

```
Enter the number of rows and columns: 3
4

Multiplication Table:
   1   2   3   4
   2   4   6   8
   3   6   9  12
```

Nested while loop (example)

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```
#include <stdio.h>
int main()
{
   int c = 1, d;
   while (c<3)
   { // Outer loop
      d=1;
      while (d<=3)
      { // Inner loop
         printf("c: %d, d: %d\n", c, d);
         d++;
      }
      printf("End of Inner Loop\n");
      c++;
   }
   return 0;
}
```

```
c: 1, d: 1
c: 1, d: 2
c: 1, d: 3
End of Inner Loop
c: 2, d: 1
c: 2, d: 2
c: 2, d: 3
End of Inner Loop
```

Nested do while loop (example)

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```
#include <stdio.h>
int main() {
   int x = 1;
   do
   { // Outer do-while loop
      int y = 1;
      do
      { // Inner do-while loop
         printf("x: %d, y: %d\n", x, y);
         y++;
      } while (y <= 4);  // Inner loop condition
      printf("End of Inner Loop %d\n",x);
      x++;
   } while (x < 3);  // Outer loop condition
   return 0;
}
```

```
x: 1, y: 1
x: 1, y: 2
x: 1, y: 3
x: 1, y: 4
End of Inner Loop 1
x: 2, y: 1
x: 2, y: 2
x: 2, y: 3
x: 2, y: 4
End of Inner Loop 2
```

Pattern Programming

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- ***Pattern programming*** *in C is a* ***classic educational exercise*** *used to master the logic of* ***nested loops*** *(****for*** *or* ***while****) and* ***condition statements****.*

The general rule of thumb for any 2D pattern is:

- ***Outer loop:*** *Controls the number of* ***rows*** *(vertical movement).*
- ***Inner loops:*** *Controls the number of* ***columns*** *printed in* ***each row****.*

For example:

- ***Star (*), Number (1 2 3), Alphabet (A B C)***
- ***Pyramid patterns***
- ***Triangle patterns***
- ***Diamond patterns***

Pattern Programming

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Simple Pyramid



Rotated Simple Pyramid



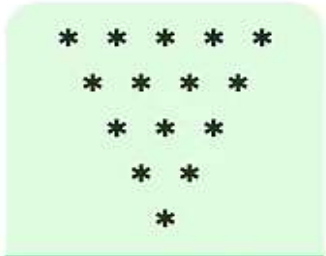
Inverted Pyramid



Rotated Inverted Pyramid



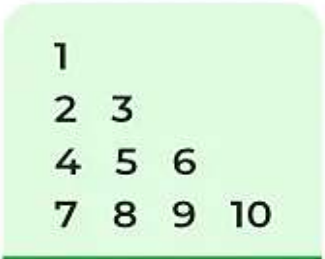
Triangle



Inverted Triangle



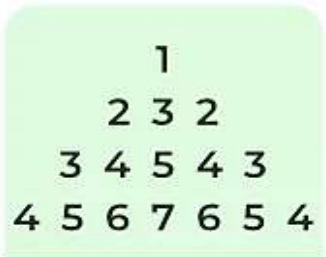
Number Pyramid



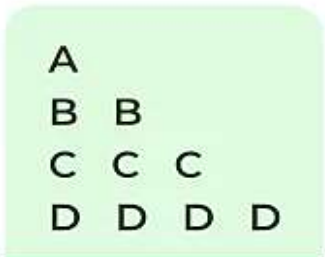
Continuous Number Pyramid



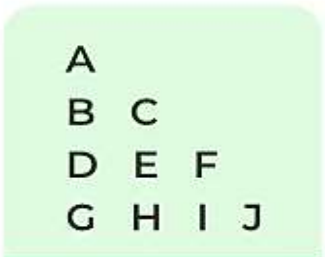
Rotated Number Pyramid



Palindrome Triangle



Alphabet Pyramid



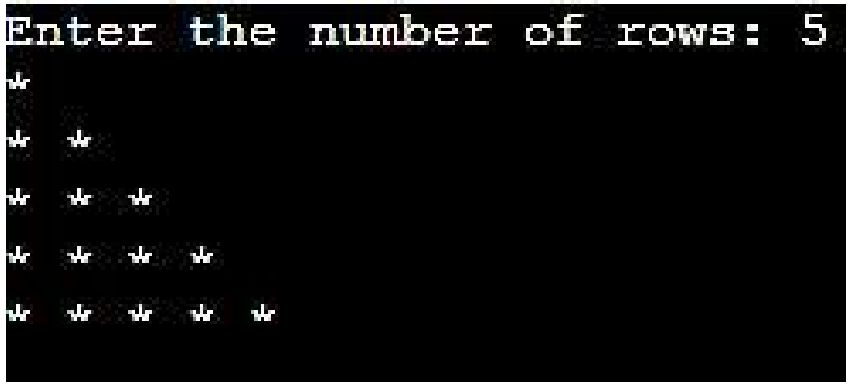
Continuous Alphabet Pyramid

Pattern Programming (examples)

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- *Pyramid of Stars(*) [Right angle triangle]*

```
#include <stdio.h>
int main()
{
        int i, j, n;
        printf("Enter the number of rows: ");
        scanf("%d", &n);
        for (i = 1; i <= n; i++)
        {
                for (j = 1; j <= i; j++)
                {
                    printf("* ");
                }
                printf("\n");
         }
         return 0;
}
```



Pattern Programming (examples)

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- *Pyramid of Stars(*) [Inverted Right angle triangle]*

```
#include <stdio.h>
int main()
{
        int i, j, n;
        printf("Enter the number of rows: ");
        scanf("%d", &n);
        for (i = n; i >= 1; i--)
        {
              for (j = 1; j <= i; j++)
              {
                  printf("* ");
              }
              printf("\n");
         }
         return 0;
}
```

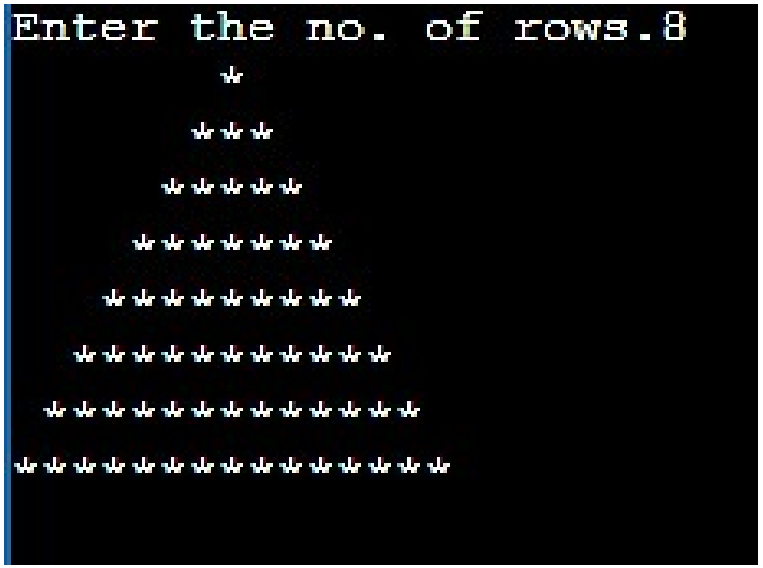
```
Enter the number of rows: 5
* * * * *
* * * *
* * *
* *
*
```

Pattern Programming (examples)

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- *Pyramid of Stars(*) [*Equilateral Triangle*]*

```
#include <stdio.h>
int main() {
   int i, j, k, rows;
   printf("Enter the no. of rows.");
   scanf("%d",&rows);
   for(i = 1; i <= rows; i++)
   {
      for(k = 1; k <= (rows - i); k++)  {
         printf(" ");    }
      for(j = 1; j <= (2*i - 1); j++)    {
         printf("*");   }
      printf("\n");
   }
   return 0;
}
```

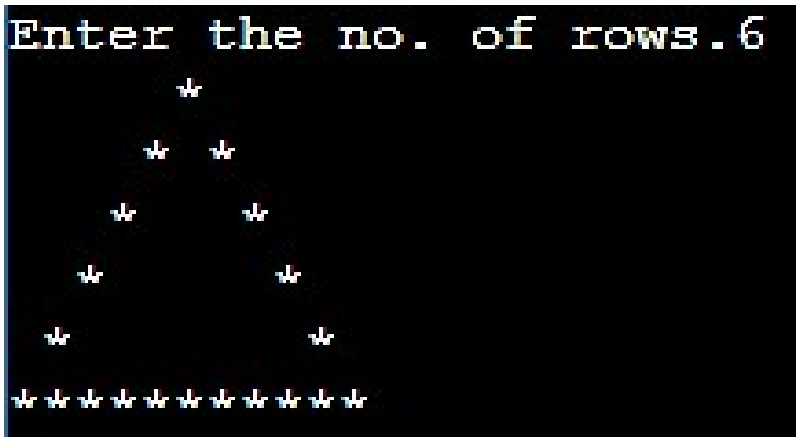


Pattern Programming (examples)

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- *Pyramid of Stars(*) [Hollow Triangle]*

```
#include <stdio.h>
int main()   {
   int i, j, rows;
   printf("Enter the no. of rows.");
   scanf("%d",&rows);
   for(i = 1; i <= rows; i++)
   {
      for(j = i; j < rows; j++)   {
         printf(" ");   }
      for(j = 1; j <= (2*i - 1); j++)  {
         if(i == rows || j == 1 || j == (2*i - 1))
            printf("*");
         else
            printf(" ");      }
      printf("\n");
   }
   return 0;   }
```



Pattern Programming (examples)

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- *Pyramid of Numbers [Right angle triangle]*

```
#include <stdio.h>
int main()
{
        int i, j, n;
        printf("Enter the number of rows: ");
        scanf("%d", &n);
        for (i = 1; i <= n; i++)
        {
                for (j = 1; j <= i; j++)
                {
                    printf("%d ", j);
                }
                printf("\n");
         }
         return 0;
}
```

```
Enter the no. of rows.5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Pattern Programming (examples)

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- *Pyramid of Numbers [Inverted Right angle triangle]*

```
#include <stdio.h>
int main()
{
        int i, j, n;
        printf("Enter the number of rows: ");
        scanf("%d", &n);
        for (i = n; i >= 1; i--)
        {
                for (j = 1; j <= i; j++)
                {
                    printf("%d ", j);
                }
                printf("\n");
         }
         return 0;
}
```

```
Enter the number of rows: 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Pattern Programming (examples)

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- *Pyramid of Alphabets [Right angle triangle]*

```
#include <stdio.h>
int main() {
   int i, j, rows;
   printf("Enter the no. of rows: ");
   scanf("%d", &rows);
   char ch;
   for(i = 1; i <= rows; i++)
   {
      ch = 'A';
      for(j = 1; j <= i; j++)   {
         printf("%c ", ch++);    }
      printf("\n");
   }
   return 0;
}
```

```
Enter the no. of rows: 5
A
A B
A B C
A B C D
A B C D E
```

Pattern Programming (examples)

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- *Pyramid of Alphabets [Inverted Right angle triangle]*

```
#include <stdio.h>
int main() {
   int i, j, rows;
   printf("Enter the no. of rows: ");
   scanf("%d", &rows);
   char ch;
   for(i = rows; i >= 1; i--)
   {
      ch = 'A';
      for(j = 1; j <= i; j++)
      {
         printf("%c ", ch++);
      }
      printf("\n");
   }
   return 0;
}
```

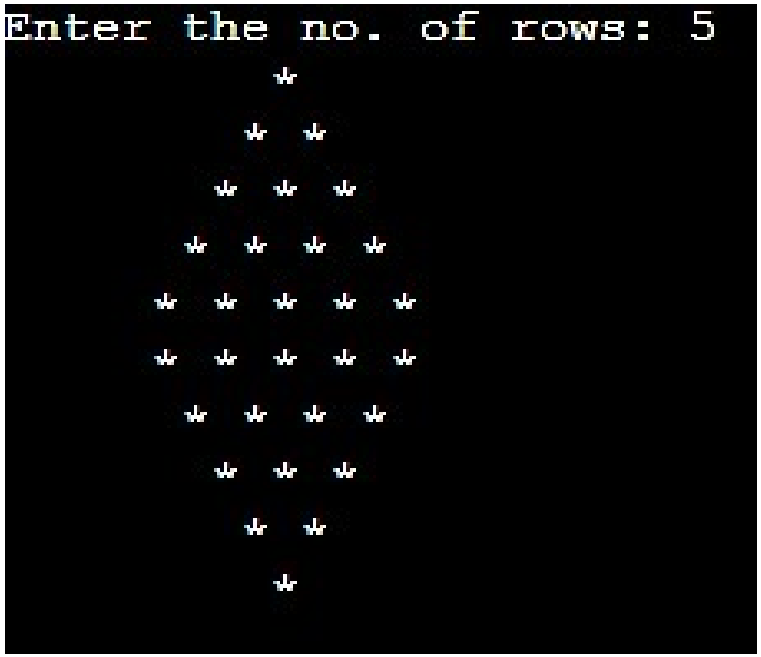
```
Enter the no. of rows: 5
A B C D E
A B C D
A B C
A B
A
```

Pattern Programming (examples)

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- *Pyramid of Numbers [Diamond Star]*

```
#include <stdio.h>
int main()    {
   int i, j, k, rows;
   printf("Enter the no. of rows: ");
   scanf("%d", &rows);
   for(i = 1; i <= rows; i++)    {
      for(k = 1; k <= 2*rows - i; k++)  {
         printf(" ");                          }
      for(j=1; j <=i; j++)    {
        printf("* ");          }
      printf("\n");                  }
   for(i = rows; i >= 1; i--)      {
      for(k = 1; k <= 2*rows - i; k++)  {
         printf(" ");                          }
      for(j=1; j <=i; j++)  {
        printf("* ");          }
      printf("\n");                  }
   return 0;     }
```



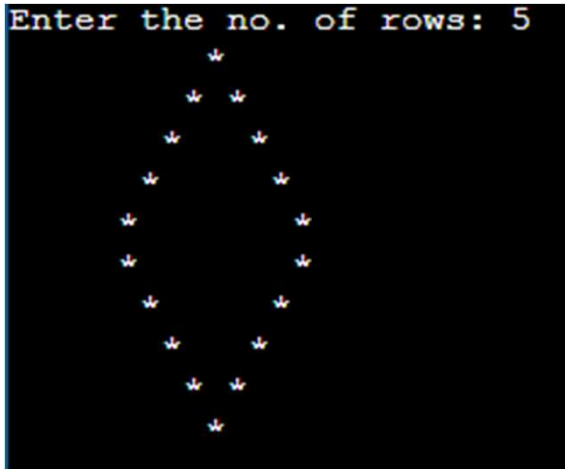
Pattern Programming (examples)

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- *Pyramid of Numbers [Hollow Diamond]*

```
#include <stdio.h>
int main() {
   int i, j, k, rows;
   printf("Enter the no. of rows: ");
   scanf("%d", &rows);
   for(i = 1; i <= rows; i++)
   {
      for(k = 1; k <= 2*rows - i; k++) {
         printf(" ");              }
      for(j = 1; j <= 2*i - 1; j++)   {
         if(j == 1 || j == 2*i - 1)
            printf("*");
         else
            printf(" ");                 }
      printf("\n");
    }
```

```
for(i = rows; i >= 1; i--)  {
      for(k = 1; k <= 2*rows - i; k++) {
         printf(" ");              }
      for(j = 1; j <= 2*i - 1; j++)   {
         if(j == 1 || j == 2*i - 1)
            printf("*");
         else
            printf(" ");                 }
      printf("\n");           }
   return 0;    }
```



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